

Date: 27 / 11 / 2025

Lab Practical #1:

- 1) Write a C program to count the number of 1's and 0's in the binary representation of a number.
- 2) Write a C program to perform bitwise AND, OR, XOR, and NOT operations on integers.
- 3) Write a C program that takes a string and performs bitwise AND and XOR operations with a constant value (e.g., 127). (Simulates simple encryption idea)
- 4) Write a C program to implement left and right circular bit rotations.
- 5) Write a C program to swap two variables without using a third variable (using XOR).

Program-1:

```
#include <stdio.h>

void countAndBinary(int n)
{
    int original = n;
    int count0 = 0, count1 = 0;
    int binary[32];
    int index = 0;

    while (n > 0)
    {
        int rem = n % 2;
        binary[index++] = rem;

        if (rem == 1)
            count1++;
        else
            count0++;

        n = n / 2;
    }

    printf("Binary of %d is:", original);
    for (int i = index - 1; i >= 0; i--)
    {
        printf("%d", binary[i]);
    }

    printf("\nCount of 1s = %d\n", count1);
    printf("Count of 0s = %d\n", count0);
}

int main()
{
    int n;

    printf("Enter any Number: ");
```



Date: 27 / 11 / 2025

```
scanf("%d", &n);  
  
countAndBinary(n);  
  
return 0;  
}
```

Output:

```
D:\CSE B-Tech\Semester-6\INS\Lab-1>gcc p1.c  
  
D:\CSE B-Tech\Semester-6\INS\Lab-1>a  
Enter any Number: 567  
Binary of 567 is: 1000110111  
Count of 1s = 6  
Count of 0s = 4
```

Date: 27 / 11 / 2025

Program-2:

```
#include <stdio.h>
int main() {
    int a, b;
    printf("Enter two integers: ");
    scanf("%d %d", &a, &b);

    printf("Bitwise AND of %d and %d is: %d\n", a, b, a & b);
    printf("Bitwise OR of %d and %d is: %d\n", a, b, a | b);
    printf("Bitwise XOR of %d and %d is: %d\n", a, b, a ^ b);
    printf("Bitwise NOT of %d is: %d\n", a, ~a);
    printf("Bitwise NOT of %d is: %d\n", b, ~b);

    return 0;
}
```

Output:

```
D:\CSE B-Tech\Semester-6\INS\Lab-1>gcc p2.c
D:\CSE B-Tech\Semester-6\INS\Lab-1>a
Enter two integers: 12
5
Bitwise AND of 12 and 5 is: 4
Bitwise OR of 12 and 5 is: 13
Bitwise XOR of 12 and 5 is: 9
Bitwise NOT of 12 is: -13
Bitwise NOT of 5 is: -6
```

Date: 27 / 11 / 2025

Program-3:

```
#include <stdio.h>
#include <string.h>

#define KEY 127

int main() {
    char str[100];
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    str[strcspn(str, "\n")] = '\0';

    int len = strlen(str);

    printf("\nOriginal String: %s\n", str);

    // AND
    printf("After AND with %d: ", KEY);
    for (int i = 0; i < len; i++) {
        printf("%c", str[i] & KEY);
    }
    printf("\n");

    // XOR (print hex)
    printf("After XOR with %d (hex): ", KEY);
    for (int i = 0; i < len; i++) {
        printf("%02X ", str[i] ^ KEY);
    }
    printf("\n");

    return 0;
}
```



Date: 27 / 11 / 2025

Output:

```
D:\CSE B-Tech\Semester-6\INS\Lab-1>gcc p3.c

D:\CSE B-Tech\Semester-6\INS\Lab-1>a
Enter a string: darshan

Original String: darshan
After AND with 127: darshan
After XOR with 127 (hex): 1B 1E 0D 0C 17 1E 11
```

Date: 27 / 11 / 2025

Program-4:

```
#include <stdio.h>

int main() {
    unsigned int num, n, left_rotate, right_rotate;
    int bits = 32;

    printf("Enter a number: ");
    scanf("%u", &num);

    printf("Enter number of positions to rotate: ");
    scanf("%u", &n);

    n %= bits;

    left_rotate = (num << n) | (num >> (bits - n));
    right_rotate = (num >> n) | (num << (bits - n));

    printf("\nOriginal Number: %u\n", num);
    printf("After Left Circular Rotation : %u\n", left_rotate);
    printf("After Right Circular Rotation: %u\n", right_rotate);

    return 0;
}
```

Output:

```
D:\CSE B-Tech\Semester-6\INS\Lab-1>gcc p4.c
D:\CSE B-Tech\Semester-6\INS\Lab-1>a
Enter a number: 45
Enter number of positions to rotate: 1

Original Number: 45
After Left Circular Rotation : 90
After Right Circular Rotation: 2147483670
```

Date: 27 / 11 / 2025

Program-5:

```
#include <stdio.h>

int main() {
    int a, b;

    printf("Enter value of a: ");
    scanf("%d", &a);

    printf("Enter value of b: ");
    scanf("%d", &b);

    // Swap using XOR
    a = a ^ b;
    b = a ^ b;
    a = a ^ b;

    printf("\nAfter swapping:\n");
    printf("a = %d\n", a);
    printf("b = %d\n", b);

    return 0;
}
```

Output:

```
D:\CSE B-Tech\Semester-6\INS\Lab-1>gcc p5.c

D:\CSE B-Tech\Semester-6\INS\Lab-1>a
Enter value of a: 23
Enter value of b: 32

After swapping:
a = 32
b = 23
```